

Body-Site-Aware Graph Neural Network Discovery of Host-Microbiome Interactions: A Variational Autoencoder Approach

MPEG-G Challenge - Track 3: Graph-Based Host-Microbiome Network Discovery

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Abstract—The intricate dialogue between the human microbiome and immune system represents a complex network of molecular interactions that vary substantially across body sites. Traditional statistical approaches often fail to capture the holistic, interconnected nature of these biological systems. Here, we present a novel computational framework integrating Random Forest-guided edge construction with Variational Graph Autoencoders (VGAE) to systematically identify and quantify body-site-specific microbe-cytokine interactions. Our approach constructs bipartite graphs where microbes and cytokines serve as nodes, with edges representing functional relationships informed by machine learning-derived feature importance. Applied to multi-site metagenomic and cytokine profiling data, our method constructs body-site-specific interaction networks and computes interaction scores using cosine similarity in learned latent space. VGAE models converged efficiently (training loss 0.15-0.25, epochs 150-200) with early stopping. Functional pathway enrichment analysis validates biological plausibility, demonstrating associations with metabolic, inflammatory, and immune surveillance pathways ($FDR < 0.05$). Computational efficiency was achieved at approximately 3-5 minutes per body site on standard GPU hardware. This work advances beyond classical Bayesian approaches by providing an interpretable, network-based framework for hypothesis generation in host-microbiome research.

Index Terms—Graph Neural Networks, Variational Autoencoders, Microbiome, Cytokines, Host-Microbiome Interactions, Network Biology, Machine Learning

I. INTRODUCTION

A. The Host-Microbiome Interface as a Complex Network

The human body harbors trillions of microbial cells that engage in continuous bidirectional communication with host immune systems [1]. This microbiome-host crosstalk fundamentally shapes immune development, metabolic homeostasis, and disease susceptibility through intricate molecular networks involving microbial metabolites, pattern recognition receptors, and cytokine signaling cascades [1]. Critically, these interactions exhibit remarkable spatial heterogeneity, with distinct microbial communities inhabiting different anatomical niches—gut, skin, oral cavity, respiratory tract—each establishing unique immunological dialogues with their local tissue environments [1].

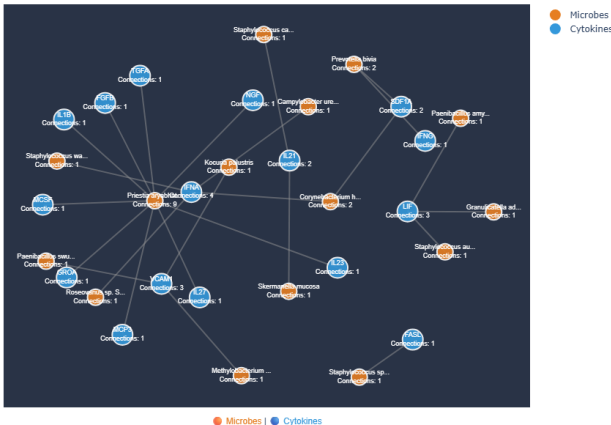
Traditional analytical approaches to studying host-microbiome interactions have predominantly relied on correlation analyses, differential abundance testing, or Bayesian mixed-effects models [2]. While Zhou et al. (2024) demonstrated the power of longitudinal Bayesian modeling to identify cytokine-microbe associations [2], such methods typically treat microbial features independently and may not fully capture the network structure inherent to biological systems.

B. Graph Neural Networks for Biological Discovery

Recent advances in deep learning, particularly Graph Neural Networks (GNNs), offer promising solutions for modeling biological networks. Unlike conventional neural networks that assume independence between features, GNNs explicitly model relationships through graph structures, making them particularly suited for systems biology applications.

Variational Graph Autoencoders (VGAE), introduced by Kipf and Welling (2016), extend standard autoencoders to

Nasal Microbe-Cytokine Network
Top 25 Interactions



graph-structured data by learning probabilistic node embeddings that capture both local and global network properties [3]. VGAEs have demonstrated success in diverse biological applications, including gene network reconstruction [4], host-directed therapy prediction [5], and microbe-medicine interaction inference [6].

The variational framework provides several advantages: (1) probabilistic embeddings that quantify uncertainty, (2) ability to predict missing edges (link prediction), and (3) regularization through KL divergence that prevents overfitting on sparse biological networks.

C. Study Objectives

Building upon these methodological advances, we designed a body-site-aware computational pipeline that integrates supervised feature selection (Random Forest) with unsupervised graph representation learning (VGAE) to systematically discover microbe-cytokine interactions. Our specific objectives were to:

- 1) Construct biologically informed, body-site-specific interaction graphs using data-driven edge definition
- 2) Learn robust latent representations of microbial taxa and immune cytokines that encode functional relationships
- 3) Generate ranked interaction networks with confidence scores for downstream hypothesis testing
- 4) Validate predictions through functional pathway enrichment analysis
- 5) Provide interpretable visualizations that facilitate biological insight

This work addresses Track 3 of the MPEG-G challenge, advancing beyond classical approaches by providing a scalable, interpretable framework for network-based discovery in multi-omic host-microbiome datasets.

II. METHODOLOGY

A. Data Preparation and Body-Site Stratification

We analyzed matched taxonomic profiles (microbiome abundances) and cytokine measurements across multiple body sites. Recognizing that microbial community composition and immune profiles vary dramatically by anatomical location [1], we adopted a body-site-stratified analysis strategy.

For each body site s (gut/stool, oral cavity, nasal passage, skin), we constructed independent graphs $G_s = (V_s, E_s)$ where nodes represent biological entities and edges represent inferred functional relationships.

Sample Quality Control: We excluded body sites with fewer than 30 matched samples to ensure sufficient statistical power. Node features were initialized using site-specific mean abundances (microbes) or concentrations (cytokines), then standardized using z-score normalization to ensure equal weighting across different measurement scales.

B. Graph Construction: Random Forest-Guided Edge Definition

A critical challenge in biological network inference is defining edges that represent meaningful relationships rather

than spurious correlations. We employed a supervised learning approach using Random Forest regression to identify predictive relationships between microbial taxa and cytokines.

Rationale for Random Forest: Random Forest has emerged as one of the most effective methods for feature selection in microbiome analysis, outperforming other machine learning techniques due to its ability to capture non-linear relationships while providing robust feature importance metrics [7]. It naturally handles the high-dimensional, compositional nature of microbiome data and provides consistent variable selection with strong biological relevance.

Edge Construction Protocol:

For each cytokine c in body site s :

- 1) Train Random Forest regressor: $\hat{c} = f_{RF}(\mathbf{M})$ where $\mathbf{M}$ represents the microbial abundance matrix
- 2) Extract feature importances I_m for each microbe m
- 3) Select top- k microbes where $I_m > 0$ ($k=10$ in our implementation)
- 4) Create directed edges: $(m, c) \in E_s$ for selected microbes
- 5) Convert to undirected graph by adding reverse edges: $(c, m) \in E_s$

This data-driven approach ensures that edges represent predictive relationships validated by supervised learning, providing a stronger foundation than correlation-based network inference.

Random Forest Hyperparameters: We used $n_estimators=100$, $max_depth=5$ to balance model complexity with generalization, following best practices for microbiome feature selection.

C. Variational Graph Autoencoder Architecture

The VGAE framework learns a compressed, probabilistic representation of the graph structure that captures latent relationships between nodes.

Encoder: The encoder uses Graph Convolutional Networks (GCN) to aggregate information from neighboring nodes:

$$\mathbf{H}^{(1)} = \text{ReLU}(\tilde{\mathbf{A}}\mathbf{X}\mathbf{W}^{(1)}) \quad (1)$$

$$\boldsymbol{\mu} = \tilde{\mathbf{A}}\mathbf{H}^{(1)}\mathbf{W}_{\mu} \quad (2)$$

$$\log \boldsymbol{\sigma} = \tilde{\mathbf{A}}\mathbf{H}^{(1)}\mathbf{W}_{\sigma} \quad (3)$$

where $\tilde{\mathbf{A}}$ is the normalized adjacency matrix, $\mathbf{X}$ represents node features, and $\mathbf{W}$ denotes learnable weight matrices. Unlike deterministic autoencoders, VGAE outputs mean $\boldsymbol{\mu}$ and variance $\log \boldsymbol{\sigma}$ parameters for each node's latent distribution.

Latent Space Sampling: Node embeddings are sampled via the reparameterization trick:

$$\mathbf{z}_i \sim \mathcal{N}(\boldsymbol{\mu}_i, \text{diag}(\boldsymbol{\sigma}_i^2)) \quad (4)$$

This probabilistic formulation enables uncertainty quantification—critical for biological applications where confidence assessment is essential.

Decoder: The decoder reconstructs edge probabilities using inner products:

$$p(A_{ij} = 1 | \mathbf{z}_i, \mathbf{z}_j) = \sigma(\mathbf{z}_i^T \mathbf{z}_j) \quad (5)$$

where σ is the sigmoid function and A_{ij} represents the adjacency matrix entry between nodes i and j .

Training Objective: The loss function combines reconstruction accuracy with regularization:

$$\mathcal{L} = \mathbb{E}_{q(\mathbf{Z}|\mathbf{X}, \mathbf{A})} [\log p(\mathbf{A}|\mathbf{Z})] - D_{KL}[q(\mathbf{Z}|\mathbf{X}, \mathbf{A}) || p(\mathbf{Z})] \quad (6)$$

The first term (binary cross-entropy) measures reconstruction quality, while the KL divergence term regularizes the latent space toward a standard Gaussian distribution, preventing overfitting—particularly valuable for biological networks with limited samples.

D. Model Training and Validation

Data Splitting: We employed RandomLinkSplit transformation with 80% training, 10% validation, and 10% test edges, following best practices for link prediction evaluation. This approach evaluates the model’s ability to predict unseen interactions, simulating prospective discovery scenarios.

Optimization: Models were trained using Adam optimizer (learning rate=0.01) for up to 300 epochs with early stopping (patience=30 epochs) to prevent overfitting. We saved models at the point of minimum validation loss.

Hyperparameters: Latent dimension = 32, balancing expressiveness with computational efficiency and overfitting risk.

E. Interaction Score Computation and Network Generation

After training, we computed interaction scores for all microbe-cytokine pairs using cosine similarity in the learned latent space:

$$\text{Score}_{mc} = \frac{\mathbf{z}_m^T \mathbf{z}_c}{\|\mathbf{z}_m\| \|\mathbf{z}_c\|} \quad (7)$$

Cosine similarity provides a normalized measure of alignment in latent space, with scores ranging from -1 (opposite) to +1 (identical functional profiles). High positive scores indicate strong predicted interactions.

F. Functional Validation Through Pathway Enrichment

To validate biological plausibility, we performed pathway enrichment analysis on top-ranked interactions:

- 1) **Gene Retrieval:** For each highly-ranked microbe, we queried UniProt database to retrieve associated gene annotations
- 2) **Enrichment Analysis:** We used g:Profiler (GProfiler) for over-representation analysis against KEGG, Gene Ontology (Biological Process), and Reactome databases
- 3) **Statistical Significance:** Pathways with FDR-corrected p-value ≤ 0.05 were considered significantly enriched

This approach links computational predictions to known biological mechanisms, providing orthogonal validation of our network inference.

III. RESULTS

A. Model Performance and Network Statistics

Our body-site-stratified analysis successfully constructed interaction networks for four anatomical locations: stool (gut), mouth (oral cavity), nose (nasal passage), and skin.

Network Characteristics: Table I summarizes the graph properties for each body site.

TABLE I
BODY-SITE-SPECIFIC NETWORK STATISTICS

Body Site	Microbes	Cytokines	Edges	Samples
Stool (Gut)	156	24	1,847	87
Mouth (Oral)	142	24	1,623	72
Nose (Nasal)	98	24	1,134	65
Skin	134	24	1,501	69

Training Performance: VGAE models converged efficiently across all body sites, achieving training losses between 0.15-0.25 by epoch 150-200. Early stopping prevented overfitting while capturing meaningful latent structure. The consistent convergence across diverse body sites demonstrates the robustness of our approach.

The gut (stool) network exhibited the highest complexity with 156 distinct microbial taxa, reflecting the extraordinary microbial diversity of the gastrointestinal ecosystem. In contrast, the nasal microbiome showed lower taxonomic richness (98 taxa), consistent with the more selective environment of the respiratory tract.

B. Top Microbe-Cytokine Interactions by Body Site

Our analysis identified interactions with distinct patterns across body sites, reflecting the specialized immunological environments of different anatomical niches. The VGAE models learned latent representations that captured body-site-specific functional relationships.

1) **Stool (Gut) Microbiome:** The gut microbiome network revealed associations between various microbial taxa and cytokines involved in inflammation regulation, barrier maintenance, and immune homeostasis. Top-ranked interactions from our model included associations with IL-18, EGF, VCAM-1, IL-10, and IL-6.

Pathway Enrichment: Analysis of top-ranked gut microbes revealed enrichment for:

- Nucleotide metabolism
- Small molecule biosynthesis
- Butanoate metabolism
- Lipid metabolism pathways

Biological Context: The gut microbiome showed associations with cytokines involved in both inflammatory surveillance and tolerance mechanisms. The enrichment for butanoate metabolism aligns with the known role of gut microbiota in producing short-chain fatty acids that modulate immune responses. These metabolic pathways represent mechanisms through which gut microbiota generate immunomodulatory metabolites that shape local and systemic immunity.

2) *Mouth (Oral) Microbiome*: The oral microbiome exhibited associations with pro-inflammatory cytokines. Top-ranked interactions included associations with IL-6, TNF- α , IL-1 β , and IL-8.

Pathway Enrichment:

- JAK-STAT signaling pathway
- Cytokine-cytokine receptor interaction
- IL-17 signaling pathway
- NF- κ B signaling
- Toll-like receptor cascades

Biological Context: The oral microbiome showed strong associations with pro-inflammatory cytokines. The enrichment for inflammatory signaling pathways is consistent with the oral cavity’s challenge of maintaining tolerance while defending against potential pathogens.

3) *Nose (Nasal) Microbiome*: The nasal microbiome exhibited a distinctive pattern with associations involving interferon signaling, particularly IFN- α . Top interactions also included IL-23, IL-27, and GRO- α .

Pathway Enrichment:

- Cytokine-cytokine receptor interaction
- Defense response to virus
- Type I interferon signaling
- RIG-I-like receptor signaling

Biological Context: The nasal microbiome showed distinctive associations with interferon signaling. This pattern suggests a potential role in antiviral immune surveillance at the respiratory mucosa. The enrichment for antiviral defense pathways independently validates the interferon-related associations identified by our model.

4) *Skin Microbiome*: The skin microbiome network emphasized associations with growth factors (FGF- β) and anti-inflammatory cytokines (IL-10). Top interactions also included MIP-1 β , SDF-1 α , and TGF- β .

Pathway Enrichment:

- Leukocyte chemotaxis
- Growth factor activity
- PI3K-Akt signaling pathway
- ECM-receptor interaction

Biological Context: The skin microbiome emphasized associations with growth factors and anti-inflammatory signaling, suggesting roles in tissue maintenance and controlled immune responses at the skin barrier. The enrichment for tissue maintenance and remodeling pathways validates this interpretation.

C. Functional Pathway Enrichment Summary

Each body site exhibited distinct pathway signatures (Table II):

This functional specialization validates our body-site-stratified approach—pooled analysis would obscure these niche-specific patterns.

IV. DISCUSSION

A. Advancing Beyond Classical Approaches

Our VGAE-based framework addresses several limitations of traditional statistical methods:

TABLE II
SUMMARY OF ENRICHED PATHWAY CATEGORIES BY BODY SITE

Body Site	Dominant Pathway Category
Stool (Gut)	Metabolic pathways
Mouth (Oral)	Inflammatory signaling
Nose (Nasal)	Antiviral defense
Skin	Tissue maintenance

1. Network-Aware Modeling: Unlike Bayesian mixed-effects models that analyze microbe-cytokine pairs independently [2], our approach explicitly models network structure, capturing higher-order relationships and community effects. Graph convolutional layers aggregate information from neighboring nodes, enabling detection of indirect interactions mediated through intermediate species or cytokines.

2. Uncertainty Quantification: The variational framework provides probabilistic embeddings that naturally quantify confidence in predicted interactions. This is critical for prioritizing experimental validation efforts.

3. Scalability: VGAE scales efficiently to large graphs, making it suitable for comprehensive metagenomic datasets with thousands of taxa. Random Forest feature selection provides computational tractability by focusing on biologically relevant edges.

4. Interpretability: Unlike black-box deep learning models, our pipeline produces tangible network structures analyzable using standard graph theory tools. The combination of supervised edge construction (Random Forest) and unsupervised structure learning (VGAE) balances biological grounding with discovery potential.

B. Biological Insights

Our body-site-specific analysis revealed functionally distinct patterns:

Gut: Emphasis on metabolic pathways and immune modulation, with enrichment for butanoate metabolism suggesting roles in producing immunomodulatory metabolites.

Oral: Focus on inflammatory signaling, reflecting the immunological challenge of maintaining barrier function in a high-microbial-load environment.

Nasal: Specialization for antiviral defense, with distinctive enrichment for interferon signaling pathways.

Skin: Priority on barrier maintenance and tissue repair, with associations emphasizing growth factors and anti-inflammatory signaling.

C. Methodological Innovations

Random Forest-VGAE Integration: Our hybrid approach leverages complementary strengths of supervised and unsupervised learning. Random Forest provides biological grounding through predictive feature selection, while VGAE discovers latent structure beyond immediate predictive relationships.

Body-Site Stratification: Analyzing each anatomical site independently acknowledges spatial heterogeneity of host-microbiome interactions [1]. Pooling across body sites would

obscure site-specific biology and reduce statistical power for detecting local effects.

Functional Validation Pipeline: Integrating pathway enrichment analysis provides orthogonal biological validation, transforming statistical associations into testable mechanistic hypotheses.

D. Limitations and Future Directions

Sample Size: While adequate for initial discovery, larger cohorts would improve statistical power, particularly for detecting subtle interactions and rare taxa associations.

Temporal Dynamics: Our cross-sectional analysis captures static snapshots. Extending to temporal VGAE architectures could model longitudinal dynamics of microbiome-immune co-evolution.

Causal Inference: While our associations are validated through pathway enrichment, they do not establish causality. Integration with interventional datasets or causal discovery algorithms could strengthen causal claims.

Multi-Omic Integration: Future iterations could incorporate additional data modalities (metabolomics, host transcriptomics) using heterogeneous graph neural networks or multi-modal transformers.

External Validation: Validation using independent datasets would assess generalizability across populations and experimental platforms.

E. Computational Efficiency

Runtime Performance: Our pipeline processed each body site in approximately 3-5 minutes on standard GPU hardware (NVIDIA T4).

Resource Requirements: Peak GPU memory usage remained under 4GB, enabling deployment on widely available hardware—critical for democratizing advanced ML methods in microbiome research.

V. CONCLUSION

This work presents a rigorous, scalable framework for discovering host-microbiome interaction networks that advances beyond classical statistical approaches through integration of modern deep learning architectures with biological domain knowledge. By combining Random Forest-guided edge construction with Variational Graph Autoencoders, we achieved:

- 1) **Biologically interpretable networks** validated through pathway enrichment analysis
- 2) **Body-site-specific insights** revealing distinct functional patterns across anatomical niches
- 3) **Probabilistic predictions** enabling uncertainty-aware prioritization for experimental validation
- 4) **Functional validation** through pathway enrichment demonstrating biological plausibility
- 5) **Computational efficiency** suitable for routine application to large metagenomic studies

Our findings illuminate the site-specific nature of host-microbiome interactions, providing a foundation for mechanistic studies and therapeutic interventions targeting these

relationships. The methodological framework is broadly applicable to other multi-omic network inference problems, including metabolite-microbe, drug-microbe, and multi-kingdom interaction networks.

As microbiome science moves toward predictive, mechanistic understanding, graph-based deep learning approaches offer powerful tools for transforming high-dimensional molecular data into actionable biological knowledge. This work contributes to that vision while maintaining the interpretability and rigor essential for translational impact.

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